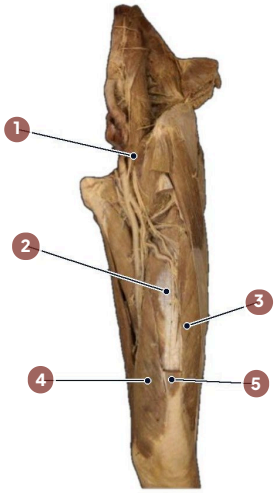
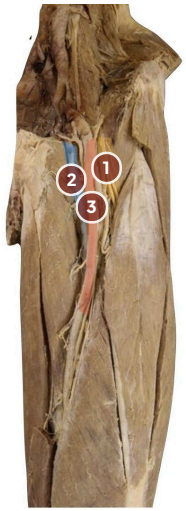
Lower Limb: Muscles, Nerves and Vessels

Competency 127 and the lab muscle list: thigh, gluteal region and leg on the cadaver, with the nerves and vessels that run with them. Name _____

A. Anterior thigh and the femoral triangle (cadaver) COMPETENCY 127, LAB



Cadaver, anterior thigh, deep



Cadaver, femoral triangle

L IN

adriceps femoris = rectus femoris, vastus lateralis, intermedius, medialis

adriceps action: extend the knee

psoas action: flex the hip

ıngle borders: inguinal ligament, sartorius, adductor longus

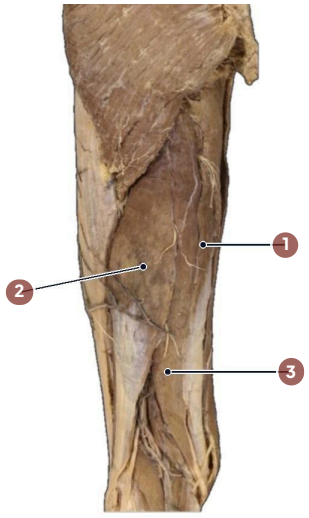
eral to medial: nerve, artery, vein (NAVEL)

noral artery was the external iliac artery

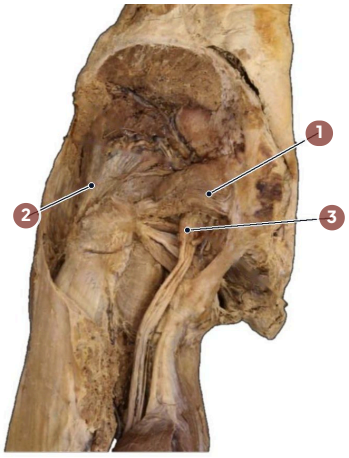
1. Iliopsoas
2. Vastus intermedius
3. Vastus lateralis
4. Vastus medialis
5. Rectus femoris

1. Femoral nerve
2. Femoral vein
3. Femoral artery

B. Gluteal region and posterior thigh (cadaver) COMPETENCY 127, LAB



Cadaver, posterior thigh: the hamstrings



Cadaver, deep gluteal region: sciatic nerve

biceps femoris, semitendinosus, semimembranosus

tions: flex the knee, extend the hip

nus: extends the hip

us: abducts the hip, steadies the pelvis

eaves the pelvis greater sciatic foramen, below piriformis

gluteus maximus, hamstrings

he knee into tibial and common fibular

1. Semimembranosus
2. Biceps femoris
3. Semitendinosus

1. Piriformis
2. Gluteus minimus
3. Sciatic nerve

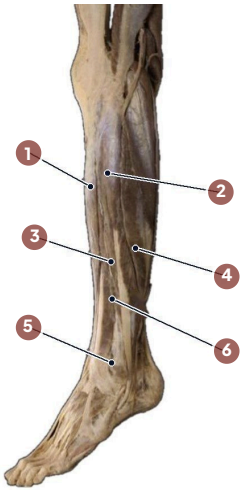
Extra notes and drawings

Extra notes and drawings

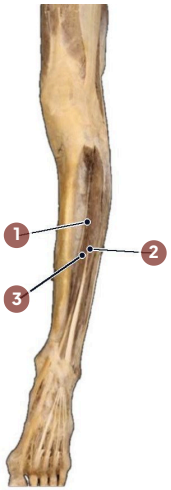
Lower Limb: Muscles, Nerves and Vessels

Competency 127 and the lab muscle list: thigh, gluteal region and leg on the cadaver, with the nerves and vessels that run with them. Name _____

C. Anterior and lateral leg (cadaver) COMPETENCY 127, LAB



Cadaver, lateral leg



Cadaver, anterior leg, deeper

1. Tibialis anterior
2. Fibularis longus
3. Superficial fibular nerve
4. Soleus
5. Extensor digitorum longus
6. Fibularis brevis

1. Anterior tibial artery
2. Extensor digitorum longus
3. Deep fibular nerve

FILL IN

Anterior leg muscles: dorsiflex, extend the toes

Fibularis longus and brevis: evert the foot

Common fibular nerve wraps the: neck of the fibula

Deep branch runs the: anterior leg

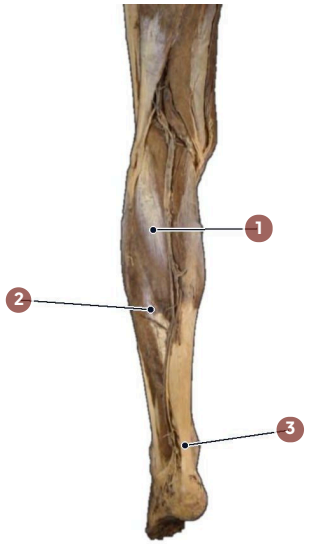
Superficial branch runs the: lateral leg

Injury at the fibular neck: foot drop

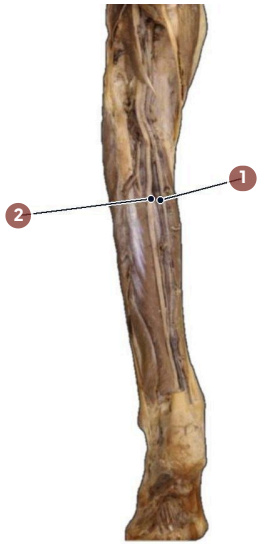
Anterior tibial artery becomes: dorsalis pedis

Extra notes and drawings

D. Posterior leg, and the veins (cadaver) COMPETENCY 127, LAB



Cadaver, posterior leg



Cadaver, deep posterior leg

1. Gastrocnemius
2. Soleus
3. Calcaneal tendon

1. Posterior tibial artery
2. Tibial nerve

nemius and soleus: plantar flex the foot

re the: calcaneal (Achilles) tendon

ne medial malleolus: posterior tibial artery, tibial nerve

ns run with the: arteries, same names

phenous vein: superficial, medial, in front of the medial malleolus, into the femoral vein

phenous vein: up the calf, into the popliteal vein

named at: inguinal ligament, adductor hiatus, below the knee

Extra notes and drawings

Lymphatic System

Competencies 105 to 108: the route lymph takes, vessel and node structure, primary and secondary organs, disorders.

Name _____

A. The route lymph takes (draw it) and the spleen (cadaver) COMPETENCY 105, 107

ONE-WAY ROUTE, IN ORDER

1. lymphatic capillaries (blind ends in the tissue)
2. collecting vessels, with valves
3. lymph nodes
4. lymphatic trunks
5. right lymphatic duct or thoracic duct
6. subclavian veins

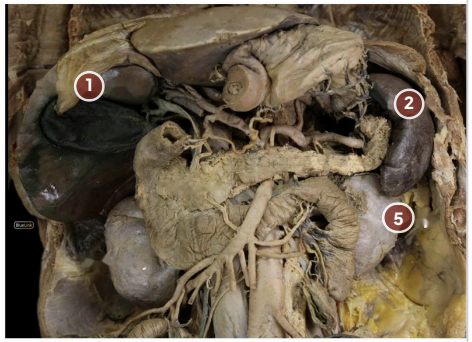
FILL IN

Lymph is: fluid the capillaries left in the tissue, picked up by lymphatic vessels

Right lymphatic duct drains: the right arm, right head and chest (upper right quarter)

Thoracic duct drains: everything else; starts at the cisterna chyli

Lymph moves with no pump by: skeletal muscle, breathing, valves



Cadaver, upper abdomen: find the spleen

1. Right lobe of liver
2. Spleen
5. Left kidney

SPLEEN

Sits: upper left, behind the stomach, under ribs 9 to 11

White pulp: lymphocytes, immune

Red pulp: blood filtering, old red cells removed

B. Lymphatic capillary, collecting vessel, lymph node (draw it) COMPETENCY 106

LYMPH NODE, OUTSIDE IN

Capsule: dense connective tissue

Cortex: lymphatic nodules, lymphocytes

Medulla: medullary cords and sinuses

Afferent vessels enter: the convex side, many

Efferent vessels leave: the hilum, few

Hilum: artery in, vein and efferent vessels out

CAPILLARY AND VESSEL

Lymphatic capillary wall: overlapping endothelial flaps, one-way minivalves

Collecting vessels have: valves, so flow is one way

Lacteals are lymphatic capillaries in: the villi of the small intestine

DRAW: ONE LYMPH NODE WITH CAPSULE, CORTEX, MEDULLA, AFFERENT VESSELS ENTERING AND EFFERENT VESSELS LEAVING THE HILUM

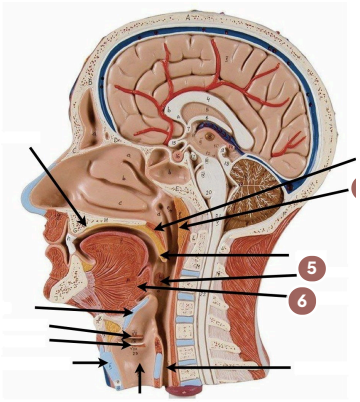
Extra notes and drawings

Extra notes and drawings

Lymphatic System

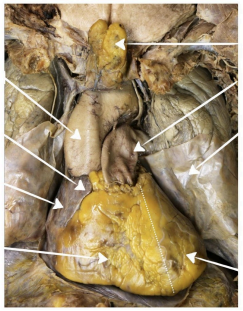
Competencies 105 to 108: the route lymph takes, vessel and node structure, primary and secondary organs, disorders. Name _____

C. Primary and secondary organs COMPETENCY 107



Model, midsagittal head: find the three tonsils

3 Pharyngeal tonsil
5 Palatine tonsil
6 Lingual tonsil



adaver, chest opened: find the thymus

Thymus

FILL IN

Primary organs: red bone marrow, thymus (lymphocytes made, mature)

Secondary organs: nodes, spleen, tonsils, MALT (where they work)

Thymus: mediastinum, behind the sternum; big in children

Pharyngeal tonsil: nasopharynx

Palatine tonsils: oropharynx

Lingual tonsil: root of the tongue

MALT: Peyer patches, appendix

Extra notes and drawings

D. Disorders: which structure does each one affect?

COMPETENCY 108

DISORDER: STRUCTURE, WHAT CHANGES

Edema: tissue fluid builds up, any cause

Lymphedema: lymph vessels blocked or removed, limb swells

Lymphadenopathy: lymph nodes enlarged, infection or cancer

Lymphoma: cancer of the lymphocytes in lymphatic tissue

Splenomegaly: spleen enlarged; rupture risk in contact sports

Tonsillitis: palatine tonsils inflamed

DRAW: THE BODY IN OUTLINE, THE THORACIC DUCT AND RIGHT LYMPHATIC DUCT, AND SHADE THE QUARTER THE RIGHT DUCT DRAINS

Extra notes and drawings